

$P \nRightarrow Q$ [P is not necessary for Q to hold]

if P then Q, P only if Q

$P \Leftrightarrow Q$

$$(P \Rightarrow Q) \wedge (Q \Rightarrow P)$$

[P is necessary & sufficient condition for Q]

P if and only if Q

$$x+3=5 \nrightarrow 0=0$$

$$\nrightarrow x+2=4$$

learned symbols:

$\wedge$

$\vee$

$\neg$

$\Rightarrow$

$\Leftrightarrow$

$$P \Leftrightarrow Q \quad (P \Rightarrow Q) \wedge (Q \Rightarrow P)$$

$$(P \vee Q) \wedge \neg (P \wedge Q)$$

Truth table:

P	Q	$P \vee Q$	$P \wedge Q$	$\neg (P \wedge Q)$	$(P \vee Q) \wedge \neg (P \wedge Q)$
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The product of $xy=0$ if and only if ($x=0$ or $y=0$)

$$\underbrace{xy=0}_P \iff \{ \underbrace{(x=0)}_Q \vee \underbrace{(y=0)}_R \}$$

P Q R Q ∨ R P ⇔ (Q ∨ R)

T T T

T

T

T T F

T

T

T F T

T

T

T F F

F

F

F T T

T

F

F T F

T

F

F F T

T

F

F F F

F

T

$\neg (P \implies Q)$

$P \wedge \neg Q$

$\uparrow$	Q	$P \Rightarrow Q$	$\sim (P \Rightarrow Q)$
$\top$	$\top$	$\top$	F
$\top$	F	F	$\top$
F	$\top$	$\top$	F
F	F	$\top$	F

P	Q	$\sim Q$	$P \wedge \sim Q$
$\top$	$\top$	F	F
$\top$	F	$\top$	$\top$
F	$\top$	F	F
F	F	$\top$	F

$$\sim (P \Rightarrow Q) = (P \wedge \sim Q)$$

$$(P \Rightarrow Q) = (\sim Q) \Rightarrow (\sim P)$$

P	Q	$\sim P$	$\sim Q$	$(\sim Q) \Rightarrow (\sim P)$
$\top$	$\top$	F	F	$\top$
$\top$	F	F	$\top$	F

F	T	T	F	T
F	F	T	T	T

$$P \Rightarrow Q$$

$$(\neg Q) \Rightarrow (\neg P)$$

De Morgan:

$$\neg (P \vee Q) = (\neg P) \wedge (\neg Q)$$

$$\neg (P \wedge Q) = (\neg P) \vee (\neg Q)$$

Commutativity:

$$P \vee Q = Q \vee P, \quad P \wedge Q = Q \wedge P$$

Distributive:

$$P \wedge (Q \vee R) = (P \wedge Q) \vee (P \wedge R)$$

$$P \vee (Q \wedge R) = (P \vee Q) \wedge (P \vee R)$$

Associative:

$$(P \vee Q) \vee R = P \vee (Q \vee R)$$

$$X \nrightarrow (P \vee Q) \wedge R \neq P \vee (Q \wedge R)$$

show:

$$(P \Rightarrow Q) \vee R = \neg((P \wedge \neg Q) \wedge \neg R)$$

without truth table?

$$R.H.S = \neg((P \wedge \neg Q) \wedge \neg R)$$

$$= \neg(P \wedge \neg Q) \vee \neg(\neg R)$$

$$\downarrow$$
$$\neg(P \Rightarrow Q)$$

$$= \neg(\neg(P \Rightarrow Q)) \vee \neg(\neg R)$$

$$= (P \Rightarrow Q) \vee R$$

$$= L.H.S$$

symbols:

$\wedge, \vee, \Rightarrow, \neg, \neg$

$$X = \{x_1, x_2, x_3, \dots, x_n\}$$

$\mathcal{B}_1$: X is a set of even numbers

$$\bullet P(x_1) \wedge P(x_2) \wedge \dots \wedge P(x_n)$$

$P(x)$: x is even

At least one element of X is even.

$$P(x_1) \vee P(x_2) \vee P(x_3) \vee \dots \vee P(x_n)$$

rather we say:

there exist $x_i \in X$ such that, $P(x_i)$ [x_i is even]

$$\exists x_i \in X, P(x_i) \text{ is even}$$

$\exists$: there exists: existential quantifier

B_1 : X is a set of even numbers

$$\bullet P(x_1) \wedge P(x_2) \wedge \dots \wedge P(x_n)$$

for all $x_i \in X$, x_i is even
 $P(x_i)$

$$\forall x \in X$$

$\forall$: for all, every, for every

$\forall$: universal quantifier

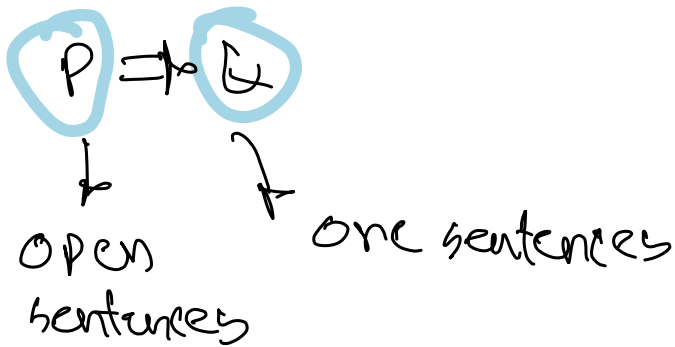
for every real x , there is ^{a real} number y
such that, $y^3 = x$

$\forall x \in \mathbb{R}, \exists y \in \mathbb{R}$ such that, $y^3 = x$

• $\exists m \in \mathbb{I}, \forall n \in \mathbb{Z}, m = n + 5$

• $\forall n \in \mathbb{N}, \exists x \in P(\mathbb{N}), |x| < n$

Conditional statement:



x is even $\Rightarrow x$ is a multiple of 6

$P(x) \Rightarrow Q(x)$

Defⁿ: if P then Q ($P \Rightarrow Q$)

hold true for open sentences as well.

Negating statement is:

R : a is a real number

$\neg R$: a is not real number

$P \vee Q \quad \neg(P \vee Q)$

$(\neg P) \wedge (\neg Q)$

$A: \underline{x \text{ is odd}} \text{ and } \underline{y \text{ is odd}}$

$\neg A: x \text{ is not odd and } y \text{ is not odd}$

$A: \text{both } x \text{ and } y \text{ are even}$

Open sentences: quantifiers:

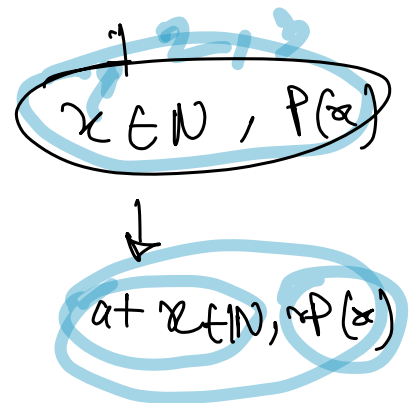
$\forall x \in \mathbb{N}, P(x)$

$\neg(\text{for every } x \in \mathbb{N}, P(x))$

for not every $x \in \mathbb{N}, \neg P(x)$

✓ at least one $x \in \mathbb{N}, \boxed{\neg P(x)}$ is TRUE

✓ $\exists x \in \mathbb{N}, \boxed{\neg P(x)}$ is TRUE



any, $\forall x \in \mathbb{R}, x^2 \geq 0$

$\neg(\forall x \in \mathbb{R}, x^2 \geq 0)$

$= \exists x \in \mathbb{R}, x^2 < 0$

there exists $x \in \mathbb{Q}$ with $x^2 = 2$

$$\neg \Rightarrow \forall x \in \mathbb{Q}, x^2 \neq 2$$

For every prime p , there is another prime q
with, $p < q$

$$\forall p \in \mathbb{P}, \exists q \in \mathbb{P} \text{ such that, } p < q$$

$$\neg \Rightarrow \exists p \in \mathbb{P}, \forall q \in \mathbb{P} \text{ with } p \geq q$$

negating with 'implication arrow':

$$R: \forall x \in \mathbb{Z} (x \text{ odd}) \Rightarrow (x^2 \text{ is odd})$$

$$\neg (\forall x \in \mathbb{Z}, \{ (x \text{ odd}) \Rightarrow (x^2 \text{ is odd}) \})$$

$$= \exists x \in \mathbb{Z}, \neg \{ (x \text{ odd}) \Rightarrow (x^2 \text{ is odd}) \}$$

$$= \exists x \in \mathbb{Z}, (x \text{ odd}) \wedge \neg (x^2 \text{ is odd})$$

"For every positive number ϵ , there is a positive number δ , such that,

$$|x - a| < \delta \Rightarrow |f(x) - f(a)| < \epsilon$$

$$\forall \epsilon > 0, \exists \delta > 0 \text{ with } |x-a| < \delta \Rightarrow |f(x) - f(a)| < \epsilon$$

$$\neg: \exists \epsilon > 0, \forall \delta > 0, \text{ with } |x-a| < \delta \text{ and } |f(x) - f(a)| \geq \epsilon$$

- For every positive number ϵ , there is a positive M , for which,

$$|f(x) - f(b)| < \epsilon \text{ whenever } x > M$$

- $\forall \epsilon > 0, \exists M > 0$, for which

$$|f(x) - f(b)| < \epsilon \text{ whenever } x > M$$

$$\neg: \exists \epsilon > 0, \forall M > 0, |f(x) - f(b)| \geq \epsilon \text{ whenever } x > M$$